THE DATA ENGINEER LAB

DATABRICKS FROM ZERO TO HERO

A 30-Day Interview & Career Readiness Series

DAY 18

Databricks SQL & SQL Warehouses

SECTION 01

What Is Databricks SQL

Databricks SQL is an analyst-friendly interface within Databricks for running SQL queries, building dashboards, and exploring data directly on the Lakehouse, without needing to write PySpark code or manage a general-purpose cluster.

CORE CONCEPT

A SQL warehouse is a compute resource optimized for running SQL queries on data stored in your Databricks lakehouse. It handles the actual query execution behind Databricks SQL, similar in spirit to a cluster but purpose-built and tuned specifically for SQL workloads.

SECTION 02

The Three SQL Warehouse Types

Databricks SQL supports three warehouse types, differing in where the compute runs and which advanced performance features are available.

Type	Compute Location	Startup Time
Serverless	Databricks-managed account	Seconds
Pro	Your own cloud account	Around 4 minutes
Classic	Your own cloud account	Around 4 minutes

SECTION 03

Performance Feature Comparison

Feature	Serverless	Pro	Classic
Photon Engine	Yes	Yes	Yes
Predictive I/O	Yes	Yes	No
Intelligent Workload Management	Yes	No	No

WHAT THESE FEATURES DO

Photon is the native vectorized query engine covered on Day 4, accelerating SQL execution. Predictive I/O optimizes how data is read from storage based on query patterns. Intelligent Workload Management dynamically and rapidly scales resources to match fluctuating query demand, which is why Serverless scales far more responsively than Pro or Classic.

SECTION 04

When to Choose Each Type

- Serverless: best for ETL, business intelligence, and exploratory analysis where fast startup and responsive autoscaling matter most; runs in the Databricks-managed account.
- Pro: a good fit when custom-defined networking is required, such as connecting to on-premises databases or private network resources, since compute runs inside your own cloud account.
- Classic: the most basic tier, still supported but offering only entry-level performance compared to Serverless or Pro.

GOOD TO KNOW

SQL warehouses do not support credential passthrough. Databricks recommends using Unity Catalog for data governance instead, tying directly back to what you learned on Day 14.

SECTION 05

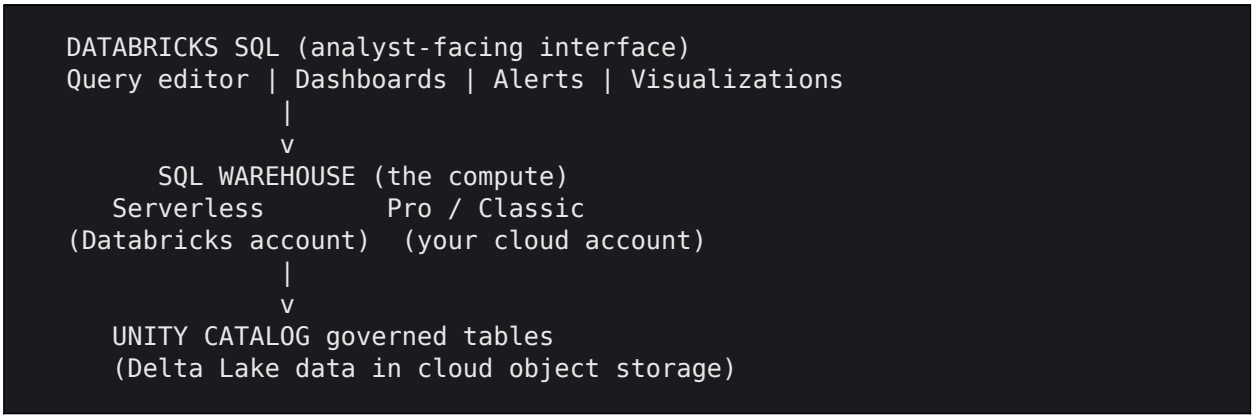
Default Warehouse Type Behavior

The default warehouse type depends on how it's created and whether the workspace region supports Serverless.

- Creating a warehouse through the UI in a region that supports Serverless defaults to Serverless.
- Creating a warehouse through the SQL Warehouses API with default parameters defaults to Classic; to get Serverless via API, you must explicitly set enable_serverless_compute to true and warehouse_type to pro.
- If the workspace uses a legacy external Hive metastore, Serverless is not supported at all, and the default falls back to Pro in the UI and Classic via the API.

SECTION 06

Where SQL Warehouses Sit in the Stack



REMEMBER THIS FOR INTERVIEWS

A SQL warehouse is conceptually similar to a cluster, it's compute, but purpose-built for SQL workloads and exposed through the Databricks SQL interface. The type you choose is a tradeoff between startup speed and autoscaling responsiveness versus networking control over the compute layer.

SECTION 07

Glossary — Day 18 Terms

Term	Definition
Databricks SQL	The analyst-facing interface for running SQL queries, building dashboards, and exploring data on the Lakehouse.
SQL Warehouse	A compute resource optimized specifically for running SQL queries against Lakehouse data.
Serverless SQL Warehouse	A warehouse type running in the Databricks-managed account, supporting all advanced performance features with near-instant startup.
Pro SQL Warehouse	A warehouse type running in your own cloud account, supporting Photon and Predictive I/O but not Intelligent Workload Management.
Classic SQL Warehouse	The most basic warehouse type, supporting only Photon among the advanced performance features.
Intelligent Workload Management (IWM)	A Serverless-only feature that rapidly and dynamically scales resources to match fluctuating query demand.

SECTION 08

Interview Questions & Answers

Q1. What is a SQL warehouse in Databricks?

A. A compute resource optimized for running SQL queries on data stored in the Databricks Lakehouse, used specifically to power the Databricks SQL interface.

Q2. What are the three types of SQL warehouses?

A. Serverless, Pro, and Classic. They differ mainly in where the compute layer runs and which advanced performance features, like Photon, Predictive I/O, and Intelligent Workload Management, are supported.

Q3. Why does a Serverless warehouse start up so much faster than Pro or Classic?

A. Because Serverless compute runs inside the Databricks-managed account and is provisioned on demand, while Pro and Classic warehouses run in the customer's own cloud account and require provisioning cloud infrastructure, which typically takes around 4 minutes.

Q4. When might a team choose a Pro warehouse over Serverless despite the slower startup?

A. When they need custom-defined networking, such as connecting to on-premises databases or private network resources, since Pro warehouses run inside the customer's own cloud account, giving more control over networking.

Q5. Does Databricks SQL support credential passthrough for governance?

A. No. SQL warehouses do not support credential passthrough; Databricks recommends using Unity Catalog for data governance instead.

Q6. If a SQL warehouse is created via the API with default settings, what type is it by default?

A. Classic, by default via the API. To provision Serverless through the API, you must explicitly set enable_serverless_compute to true and warehouse_type to pro.

STAY ON TRACK FOR ALL 30 DAYS

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Piyush Varma | The Data Engineer Lab